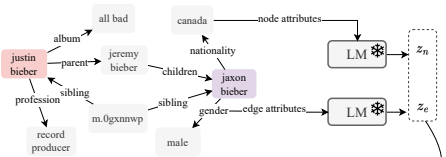


## Step 1: Indexing



Question: What is the name of justin bieber brother?

LM

$z_q$

Storage

$$V_k = \text{argtopk}_{n \in V} \cos(z_q, z_n)$$

justin bieber, this is justin bieber, jeremy bieber, justin bieber fan club, justin ...

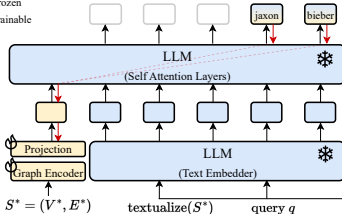
$$E_k = \text{argtopk}_{e \in E} \cos(z_q, z_e)$$

sibling, sibling\_s, hangout, friendship, friend ...

## Step 2: Retrieval

## Step 4: Generation

❄️ frozen  
🔥 trainable



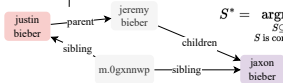
node\_id, node\_attr  
15, justin bieber  
294, jaxon bieber  
356, jeremy bieber  
551, m.0gxnnwp  
src, edge\_attr, dst  
294, parents, 356  
356, children, 15  
551, sibling, 294  
551, sibling, 15

Question: What is the name of justin bieber brother?  
Answer:

$S^* = (V^*, E^*)$

textualize( $S^*$ )

query  $q$



$$S^* = \underset{\substack{S \subseteq G, \\ S \text{ is connected}}}{\text{argmax}} \sum_{n \in V_S} \text{prize}(n) + \sum_{e \in E_S} \text{prize}(e) - \text{cost}(S)$$

## Step 3: Subgraph Construction